CS4040/5040 Project (Phase 1)

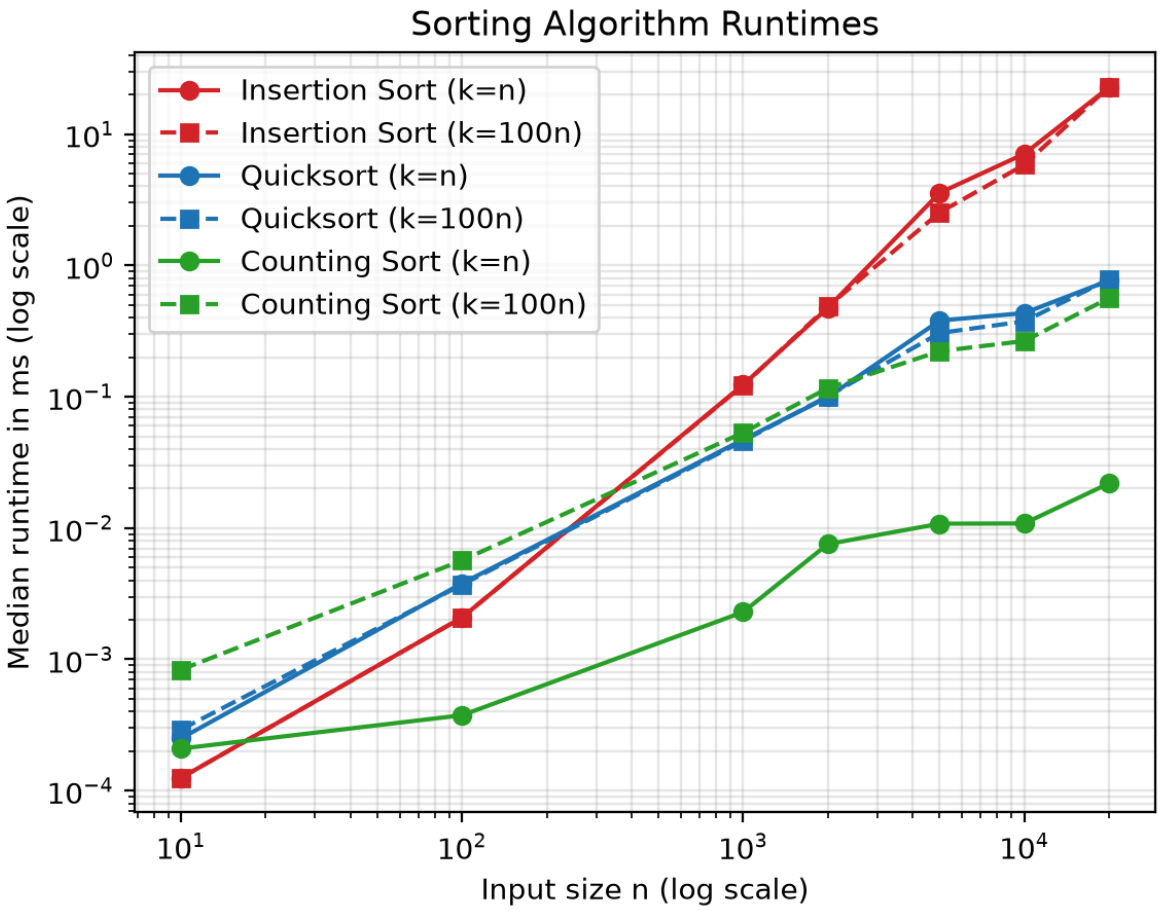
Empirical Analysis of Sorting Algorithms

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Table of Median Runtimes (built in results.csv):

n	k	InsertionSort(ms)	QuickSort(ms)	CountingSort(ms)
10	10	0.000125	0.00025	0.000209
10	1000	0.000125	0.000292	0.000834
100	100	0.002084	0.003791	0.000375
100	10000	0.002083	0.003708	0.005666
1000	1000	0.1235	0.047166	0.002292
1000	100000	0.122583	0.046125	0.053333
2000	2000	0.47525	0.09925	0.007542
2000	200000	0.48625	0.100959	0.116291
5000	5000	3.57392	0.380166	0.010792
5000	500000	2.53121	0.305833	0.22225
10000	10000	7.07904	0.434125	0.010834
10000	1000000	5.84492	0.375208	0.265792
20000	20000	22.6865	0.767334	0.021875
20000	2000000	22.8723	0.78275	0.56325

Plot (plot.png):



Discussion:
Insertion Sort gets slow pretty quickly. For n = 20,000, it takes about 22.7 ms, while Quicksort takes less than 1 ms. When n doubles from 10,000 to 20,000, Insertion Sort takes about 3.2 times longer, which is close to the 4 times expected from quadratic time. Quicksort takes about 2 times longer (0.43 to 0.77 ms), which fits its expected average O(n log n) growth.

Counting Sort grows about 9.5 times from n = 1,000 to 20,000 when k = n, which is less than the 20 times growth in n. This is consistent with its expected O(n + k) complexity. Changing k does not matter for Insertion Sort or Quicksort because they do not use k, so small differences, like 3.57 to 2.53 ms at n = 5,000, likely come from the different random inputs. Changing k matters much more for Counting Sort. At n = 20,000, using k = 100n takes about 25.75 times longer because the count array is much larger, even though both settings have O(n) complexity when k is proportional to n.

The fastest algorithm depends on n and k. For n = 10, Insertion Sort is fastest for both k values. When k = n, Counting Sort is fastest starting at n = 100. When k = 100n, Insertion Sort is fastest at n = 100, Quicksort is fastest at n = 1,000 and 2,000, and Counting Sort is fastest starting at n = 5,000. The n = 10 times are very small and potentially noisy. Counting Sort with k = n also has an unusual flat spot at n = 5,000 and 10,000, which may be caused by measurement variation.